Import Data:

I imported `List of Orders.csv` into Power BI using Home → Get Data → Text/CSV and clicked Transform Data to open Power Query Editor. Then, I used Home → New Source → Text/CSV to import `Order Details.csv` and `Sales target.csv` into Power Query Editor.

Data Transformation:

● Restrict the "List of Orders" table to only the first 500 rows.

In Power Query Editor, select the List of Orders table, go to Home → Keep Rows → Keep Top Rows, enter 500, and click OK. This restricts the table to only the first 500 rows.

● Ensure the “Order Date” column in the “List of Orders” table is set to data type 'Date'.

In Power Query Editor, I selected the List of Orders table and clicked the Order Date column. Then, I went to the Transform tab, selected Data Type, chose Date, and confirmed the change to set the column's data type to Date.

● Change the data type of “Amount” and “Target” columns to ‘Fixed Decimal Number’.

In Power Query Editor, I selected the Amount column in the List of Orders table and changed its data type to Fixed Decimal Number using Transform → Data Type → Fixed Decimal Number. I repeated the same steps for the Target column in the Sales target table.

● Format the "Customer Name" column into proper case, ensuring consistent capitalization for each word.

In Power Query Editor, I selected the Customer Name column in the List of Orders table, went to the Transform tab, clicked Format → Capitalize Each Word to convert the first letter of each word to uppercase, ensuring consistent capitalization throughout the column.

● Merge the "State" and "City" columns to create a new column named "Location" in the format ‘City, State’.

In Power Query Editor, I selected the State and City columns by holding Ctrl and clicking each column header. Then, I went to Transform → Merge Columns, selected a comma followed by a space (`, `) as the separator, and entered Location as the new column name. Finally, I clicked OK to create the column in the format City, State.

● Create a new custom column named "Profit Margin" as the percentage of "Profit" divided by "Amount".

In Power Query Editor, I selected the List of Orders table and went to Add Column → Custom Column. I entered Profit Margin as the new column name and used the formula `[Profit] / [Amount]`. Then, I clicked OK to create the column and changed its data type to Percentage using Transform → Data Type → Percentage.

● Add a new conditional column named "Profit Status" based on the values in the "Profit" column. The conditions are as follows: if the profit is less than 0, the label should be "Loss"; if the profit equals 0, the label should be "Break-Even"; and if the profit is greater than 0, the label should be "Profit".

In Power Query Editor, I selected the List of Orders table and went to Add Column → Conditional Column. I entered Profit Status as the new column name, set the first condition to Profit is less than 0, and assigned the output Loss. I added two more conditions: Profit equals 0 with the output Break-Even, and Profit is greater than 0 with the output Profit. Finally, I clicked OK to create the conditional column.

Merging Data (Joins):

● Merge the "List of Orders" and "Order Details" tables into a new single table named "Orders Data" based on the "Order ID" relationship.

In Power Query Editor, I selected the List of Orders table and clicked Home → Merge Queries → Merge Queries as New. I selected Order Details as the second table and chose the Order ID column in both tables. Then, I selected Left Outer as the join kind, clicked OK, and renamed the new table to Orders Data.

Handling Missing Data & Duplicate Data:

● Identify missing values in the data and determine a strategy to address them.

In Power Query Editor, I enabled Enable column profile and Show column quality details under the View tab to identify missing values across all columns. I checked the valid, error, and empty percentages for each column and found that the displayed columns had no missing values, with 100% valid and 0% empty.

● Check for duplicate rows and define a strategy to handle duplicates.

In Power Query Editor, I selected the relevant table and selected all columns to check for exact duplicate rows. Then, I went to Home → Keep Rows → Keep Duplicates to display duplicate records. I reviewed the results and planned to remove only exact duplicate rows using Home → Remove Rows → Remove Duplicates, while retaining

legitimate repeated Order IDs associated with different products. If no duplicate rows were found, no removal was necessary.

Sorting and Filtering Data:

● In the ‘Orders Data’ table, utilize sorting and filtering techniques on columns like Order Date, State or Category to analyze data based on specific criteria:

◆ Sort the orders by Order Date in descending order to analyze recent trends.

In Power Query Editor, I selected the Order Date column in the Orders Data table and clicked the dropdown arrow or went to Home → Sort Descending. This sorted the orders by Order Date in descending order, displaying the most recent orders first to help analyze recent trends.

◆ Filter the orders to focus only on a specific state (e.g., Tamil Nadu) for regional analysis.

In Power Query Editor, I selected the State column in the Orders Data table and clicked the filter dropdown. I selected Tamil Nadu from the list of states and clicked OK to display only orders from Tamil Nadu for regional analysis.

Grouping and Aggregating Data:

● Duplicate the “Order Details” table and calculate the count of each Order ID, average profit by Category or total amount by Sub-Category.

1. Duplicate the Order Details table: In Power Query Editor, I right-clicked the Order Details table in the Queries pane and selected Duplicate to create a copy of the table for analysis.

2. Calculate the count of each Order ID: I selected the duplicated table and went to Home → Group By. I selected Order ID as the grouping column, entered Order Count as the new column name, chose Count Rows as the operation, and clicked OK to calculate the number of records for each order.

3. Calculate average profit by Category: I opened Group By, selected Category as the grouping column, entered Average Profit as the new column name, selected Average as the operation, chose the Profit column, and clicked OK to calculate the average profit for each category.

4. Calculate total amount by Sub-Category: I opened Group By, selected Sub-Category as the grouping column, entered Total Amount as the new column name, selected Sum

as the operation, chose the Amount column, and clicked OK to calculate the total amount for each sub-category.

● Duplicate the “Sales Target” table and aggregate the total target amount by Month of Order Date.

1. Duplicate the Sales Target table: In Power Query Editor, I right-clicked the Sales target table in the Queries pane and selected Duplicate to create a copy for analysis.

2. Aggregate the total target amount by month: I selected the duplicated table and went to Home → Group By. Since the table contains Month, Category, and Target columns, I selected Month as the grouping column, entered Total Target as the new column name, chose Sum as the operation, selected the Target column, and clicked OK to calculate the total target amount for each month across categories.

Data Modeling:

● Establish a relationship between the “List of Orders” and “Order Details” tables using the ‘Order ID’ column.

I opened Model view in Power BI Desktop and located the List of Orders and Order Details tables. Then, I dragged the Order ID column from the List of Orders table onto the Order ID column in the Order Details table. I verified that the relationship was set to One-to-many (1:*), with List of Orders on the one side and Order Details on the many side, selected Single as the cross-filter direction, and clicked OK to establish the relationship.

● Build a relationship between the “Order Details” and “Sales Target” tables based on the ‘Category’ column. Click "Manage relationships" and ensure this relationship is active.

I opened Home → Manage relationships in Power BI and clicked New to create a relationship. I selected Order Details and its Category column as the first table, then selected Sales Target and its Category column as the second table. I checked the cardinality and cross-filter settings, ensured the Make this relationship active option was selected, and clicked OK to create the active relationship.